

Nandunam Sai Kumar

New Delhi | nandunamsaikumar.du.or.25@gmail.com | +91-6302041774 | linkedin.com/in/sai-kumar-nandunam

Profile Summary

Model Risk Analyst with dual MSc degrees in Operational Research and Statistics, offering strong hands-on experience in validating PD, LGD, and NMD models under CECL, CCAR, and DFAST frameworks. Proficient in SAS, Python, SQL, and Excel, with proven ability to drive regulatory compliance (SR 11-7, Basel III), automate validation workflows, and deliver robust benchmarking, sensitivity, and stability testing for Tier-1 US banking clients.

Education

Degree	Specialization	Institute	Year	CGPA/%
MSc	Operational Research	Delhi University	2023–2025	70%
MSc	Statistics	Ramjas College	2021–2023	70%
BSc (Hons)	Statistics	Ramjas College	2018–2021	8.05
HSC	Physics, Chemistry, and Maths	TTWRJC Adilabad	2018	90.70%
SSC	Science	ZPHS Edbid	2016	87

Work Experience

Anaptyss India, Model Validation Analyst (Model Risk Management) Dec 2024–Present

- Validated credit risk **LGD (Loss Given Default)** and Liquidity Risk (**Non-Maturity Deposit (NMD)** pricing models) under **CECL, CCAR, and DFAST** frameworks using **SAS and Python**, ensuring compliance with SR 11-7 and Basel III through data quality, accuracy testing, benchmarking, stability analysis (PSI/CSI), Parameter Stability, sensitivity analysis, stress testing, and scenario analysis.
- Led **Annual Model Reviews (AMR)** for **Residential Mortgage** model under CECL and CCAR, Conducting quantitative and qualitative evaluations of model methodology, risk tiering, performance monitoring, and change management to enhance risk governance.
- Utilized **SecureFX** for secure data transfers, with CI/CD integration managed via **GitLab** to enhance efficiency, accuracy, and audit readiness in credit and liquidity risk model validation.
- Applied **time series modeling** with the **Error Correction Model (ECM)** to NMD pricing and liquidity risk models, ensuring robust performance across diverse economic scenarios.
- Contributed to model documentation, ensuring clarity, transparency, and audit readiness across the model life-cycle in line with regulatory standards.
- Collaborated with model development teams to refine features and calibrate performance, leveraging proficiency in **SAS, Python, SQL, and Excel** to deliver data-driven risk solutions.

Academic Projects

Prediction of Discounted Prices on E-commerce Websites

- Developed a predictive model for discounted prices of products listed on e-commerce websites using 13 features, applying MissForest (**Random Forest Imputation**) for missing values and creating 40+ variables through feature engineering.
- Built ML pipelines and selected **Random Forest Regressor** with an **Adjusted R²** of 0.92 after evaluating multiple regression algorithms.

Credit Risk Modeling – Probability of Default (PD) Model

- Designed a **PD model** for credit card customers using application and repayment data, applying Vintage Analysis, **Weight of Evidence (WoE)** and **SMOTE** to address class imbalance, achieving a **Logistic Regression** model with a Recall of 0.74.
- Conducted feature engineering with **Information Value (IV)** to select predictive variables, enhancing model accuracy and defaulter identification.

Sentiment Analysis of Movie Reviews

- Analyzed sentiment of 6,250 IMDb movie reviews using **NLP techniques**, including **TF-IDF** vectorization and **SMOTE** for class imbalance.
- Implemented TextBlob, VADER, and lexicon-based methods, achieving a Logistic Regression accuracy of 0.85.

Technical Skills

SAS | Python | R | IBM SPSS | SQL | Excel

Statistical Modeling (Regression, Time Series, Hypothesis Testing); ML (Random forest,XGBOOST)